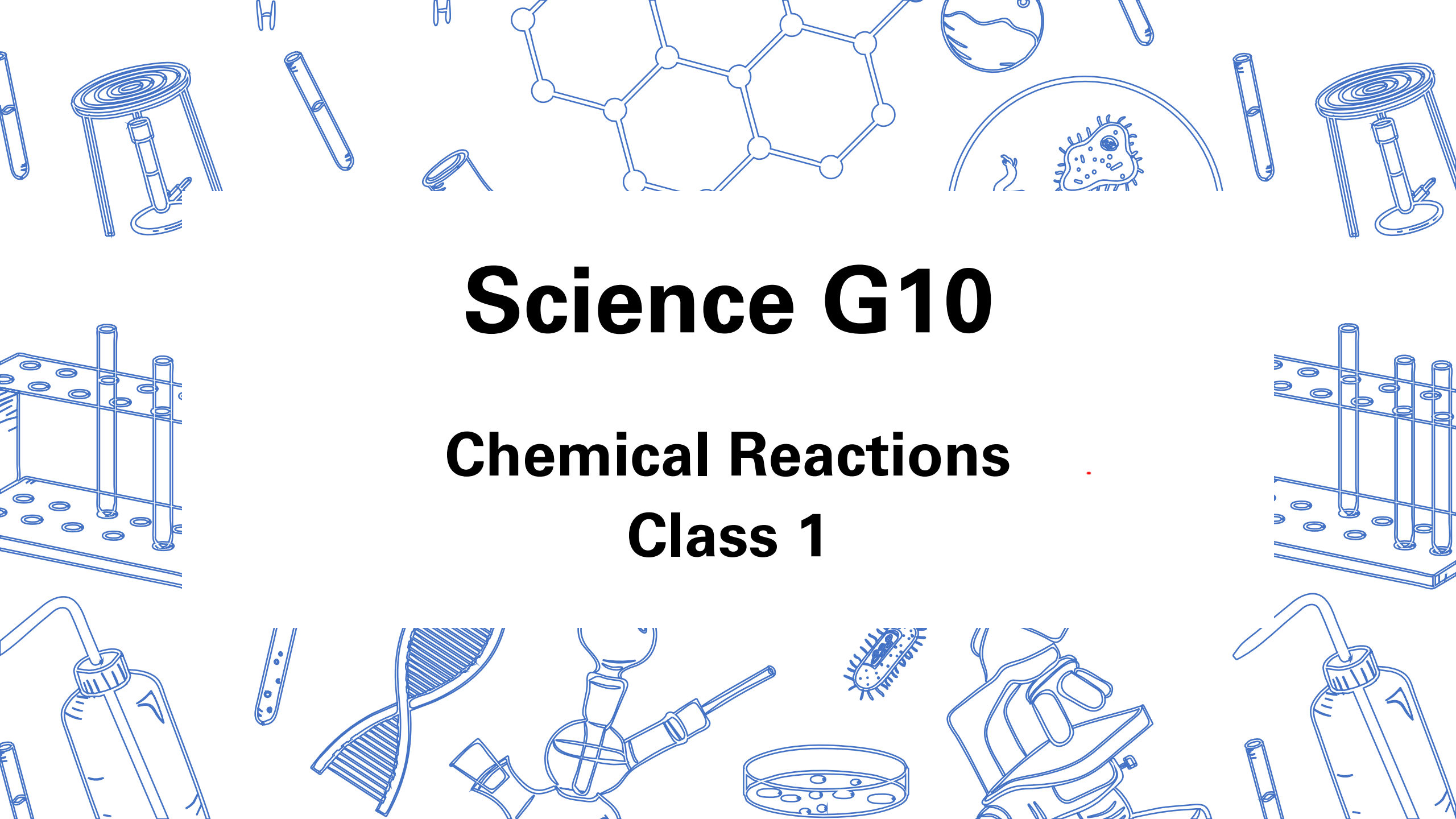


The background features a collage of blue line-art illustrations related to science. At the top center is a molecular structure with interconnected hexagons. To its right is a petri dish containing a microorganism. Below the molecular structure is a DNA double helix. To the right of the DNA is a hand holding a microscope. At the bottom center is a petri dish with several small circles representing cells. To the left of the DNA is a test tube with bubbles. To the right of the hand holding the microscope is another petri dish with cells. In the top left and right corners are illustrations of test tubes and a Bunsen burner. In the bottom left and right corners are illustrations of a test tube rack and a test tube with bubbles. The text is centered in the middle of the image.

Science G10

Chemical Reactions

Class 1

Agenda

1. Course Outline
2. Classkick Portfolio Account Setup
3. Diagnostic Test
4. Lesson 1
 - Quick Review of G9 Chemistry
 - Lewis Dot Diagrams
 - Ionic Compounds & Nomenclature
 - Covalent Bonding & Nomenclature



Chemistry

- **Chemistry** is the study of matter and the changes it undergoes
- **Matter** is anything that has mass and volume



Physical & Chemical Properties

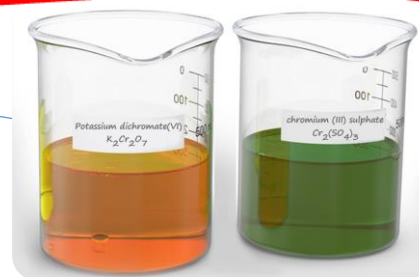
Physical Property can be measured and observed without changing the composition or identity of a substance

- Colour
- Density
- Solubility
- Melting point

Chemical Property can be observed when a substance changes into one or more new substances

- Flammability
- Bleaching
- Corrosion
- Rusting

New colour appears



Change is difficult to reverse

~~Possible~~ Evidence of Chemical Changes

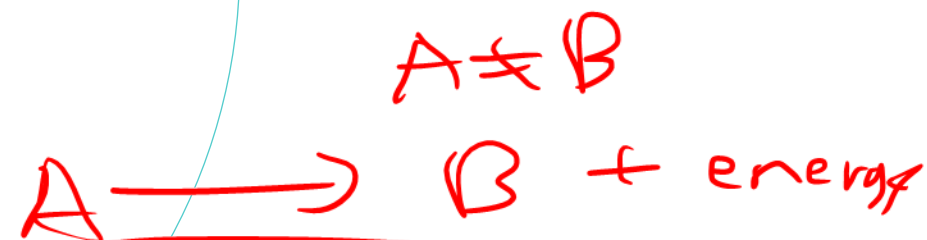
Solid material (precipitate) is formed

New solid



Bubbles or gas form

Heat or light is produced or absorbed





Checkpoint



Classify the following as a physical or chemical property:

a) Liquid nitrogen boils at -196°C

Phys

b) Propane, leaking from a tank, ignites easily

Chem

c) Silver jewelry tarnishes in air

Chem

d) Spilled oil floats on the surface of water

Phy

e) Meat darkens when it is heated on a grill

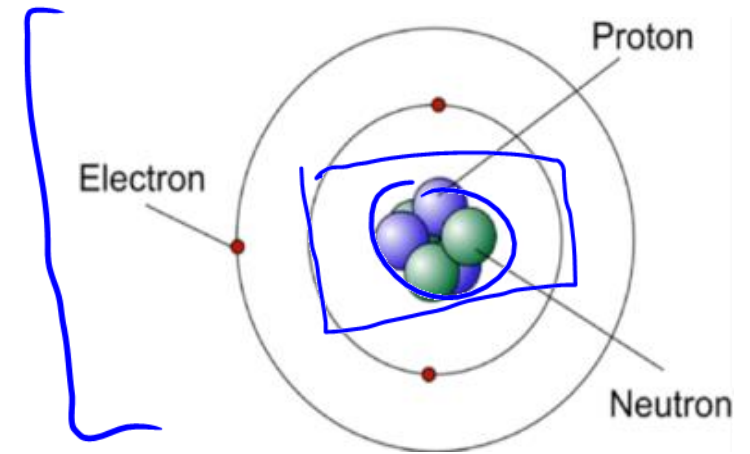
Chem

f) Sulfur trioxide changes to sulfuric acid in the atmosphere



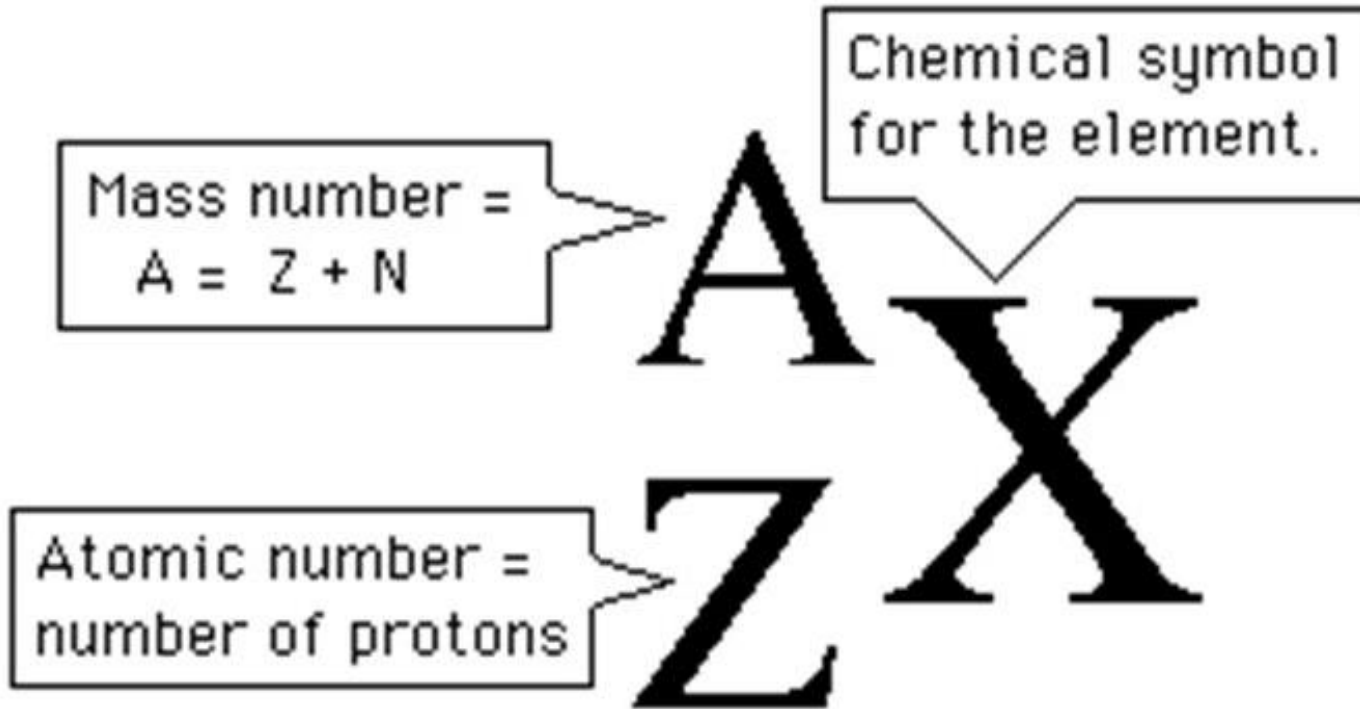
Atomic Structure

- An atom consists of:



	Proton	Neutron	Electron
Electrical Charge	<u>+</u>	0 ^{no} charge	<u>-</u>
Relative Mass	1	1	1/2000
Symbol	p ⁺	n ⁰	e ⁻
Location	Nucleus	Nucleus	Outside Nucleus

H



- **Mass Number (A)** is the total number of neutrons and protons present in the nucleus of an atom of an element
- **Atomic Number (Z)** is the number of protons in the nucleus of each atom of an element

Terms to Know

- **Isotopes** are atoms with the same atomic number but different mass numbers
- **Ions** are atoms with the same atomic number but a different number of electrons
 - **Cation** is a positively-charged ion
 - **Anion** is a negatively-charged ion
 - **Note: Atoms become ions when there is a loss/gain of electrons NOT protons**

$$10 + (-9) = +1$$

$\begin{matrix} 3 & 3 \\ p & n \end{matrix}$
 $\begin{matrix} 3 & 4 \\ (p+n) \\ \text{mass} \\ \# \end{matrix}$
 Isotopes

	Li-6	Li-7
Protons	3	3
Electrons	3	3
Neutrons	3	4

Ion

	Na	Na ⁺
Protons	11	11
Electrons	11	10
Neutrons	12	12

41
1 less electron

→ 13



Checkpoint



$$9.51 \xrightarrow{\text{rounding}} 9$$

$$12.5 \rightarrow 12$$

$$13.5 \rightarrow 14$$

List the number of p^+ , n^0 , and e^- for:

a) ⁹₄Beryllium

b) Phosphorus

c) Fluoride ion (F^-)

$$(p + n) = 9 \quad n = 9 - 4 = 5$$

$$p = 4$$

$$4p^+$$

$$5n^0$$

$$4e^-$$

$$15p^+$$

$$16n^0$$

$$15e^-$$

$$+9$$

$$x + y = -1$$

$$y = -1 - 9$$

$$9p^+$$

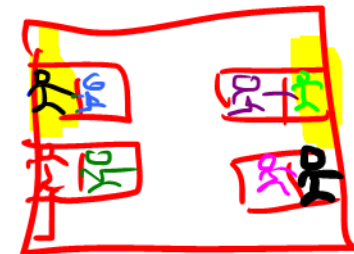
$$y = -10$$

$$10n^0$$

$$10e^-$$

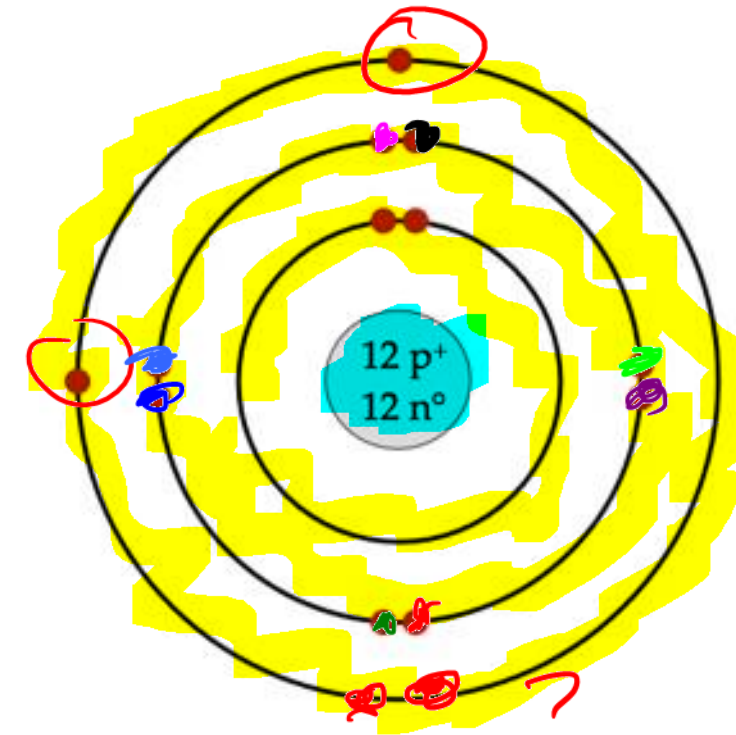
Bohr-Rutherford Diagrams

electron shells



down bus

1. Determine the number of protons, neutrons, and electrons
2. Draw the nucleus and write the number of protons and neutrons inside
3. Add the electrons to the outer shells:
 - 1st shell = 2 electrons —
 - 2nd shell = 8 electrons —
 - 3rd shell = 8 electrons — 4 pairs
 - 4th shell = 18 electrons —



Magnesium

single pair

first then



Checkpoint

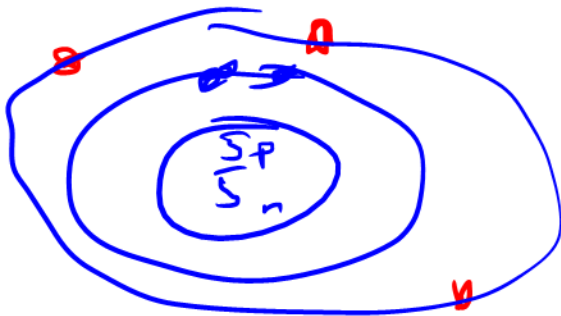


Draw the Bohr-Rutherford Diagram for:

a) Boron-10 $5e^- - 2 = \underline{3}$

$$10 = p^+ + n$$
$$5 = p$$

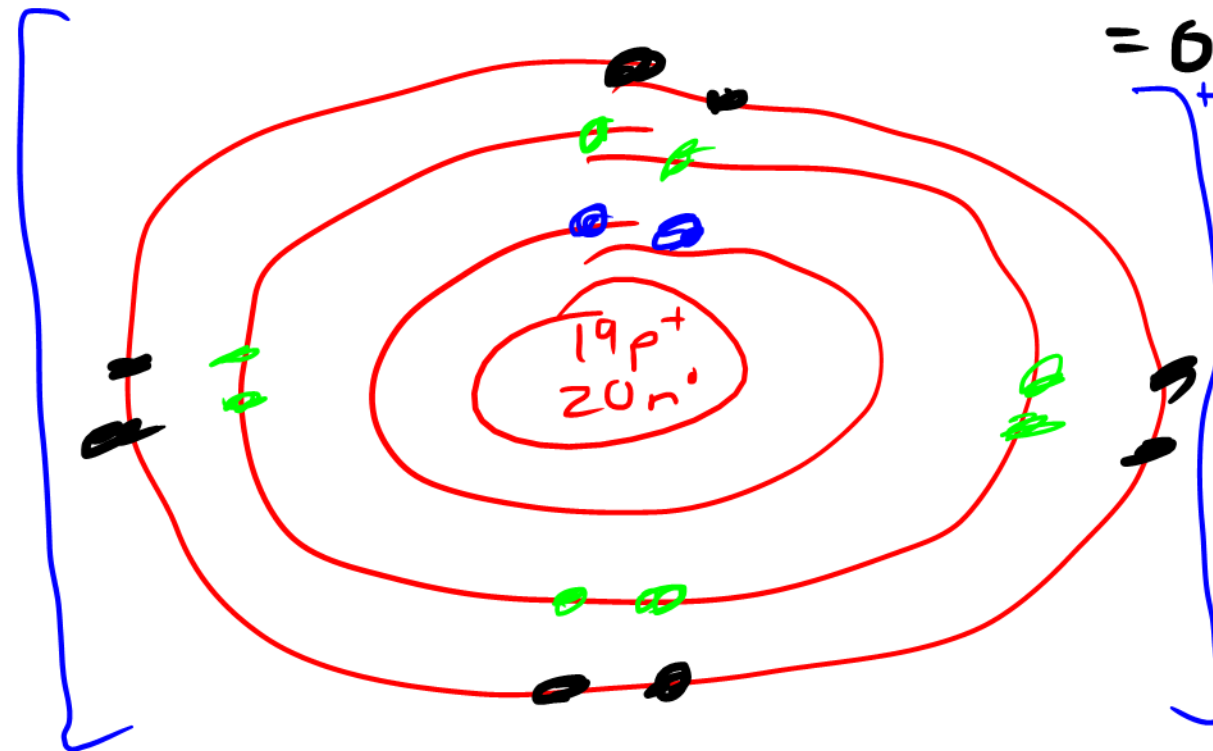
$$n = 10 - 5 = 5$$



b) K^+

$$18e^- - 2 = 16 - 8 = 8$$

$$-8 = 6^+$$



Patterns in the Periodic Table

- **Group/Family** is a column in the periodic table
- **Period** is a row in the periodic table

Diagram of the periodic table showing element symbols, atomic numbers, and names. The table is color-coded by groups and periods. Handwritten annotations include:

- metal** (blue arrow pointing to the left side of the table)
- nonmetal** (red arrow pointing to the right side of the table)
- gain e⁻** (red arrow pointing to the right side of the table)
- lose e⁻** (blue arrow pointing to the left side of the table)
- halogens** (blue arrow pointing to the right side of the table)

The periodic table is organized into groups (columns) and periods (rows). The groups are labeled with Roman numerals I through 10, and the periods are labeled with Roman numerals I through 7. The elements are color-coded by their properties:

- Alkali metals** (Group 1, red)
- Alkaline earth metals** (Group 2, orange)
- Transition metals** (Groups 3-10, blue)
- Post-transition metals** (Groups 11-12, purple)
- Metalloids** (Groups 13-14, yellow)
- Nonmetals** (Groups 15-16, green)
- Noble gases** (Group 18, pink)
- Lanthanides** (Period 6, light blue)
- Actinides** (Period 7, light green)
- Unknown chemical properties** (Groups 13-14, light blue)

	Alkali metals
	Alkaline earth metals
	Transition metals
	Post-transition metals
	Metalloids
	Nonmetals
	Noble gases
	Lanthanides
	Actinides
	Unknown chemical properties

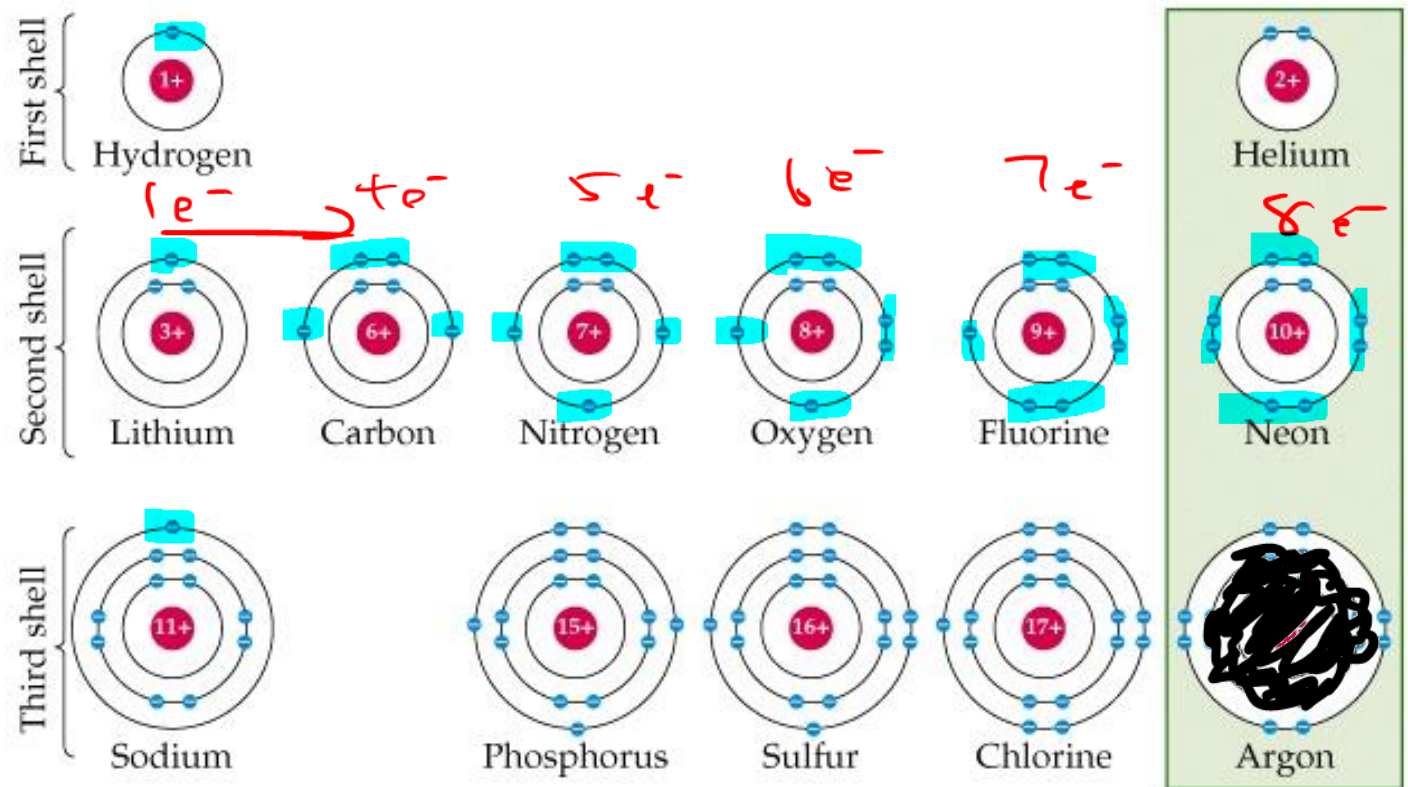
Trends in the Periodic Table

- Atomic number steadily increases across a period
- Number of energy levels (shells) steadily increases down a group
- Metals tend to lose electrons
- Nonmetals tend to attract electrons
- Number of valence electrons (outermost electrons) are the same within a group and increase steadily across a period

Reactivity

- Valence electrons determine the reactivity of an element and how compounds are formed
- Elements tend to lose or gain valence electrons to form bonds and achieve stability

Noble gases are stable due to a complete valence shell



Alkali metals are very reactive; needs to lose $1e^-$

Halogens are very reactive; needs to gain $1e^-$

Lewis Dot Diagrams

1. Write the element symbol
2. Find the number of valence electrons by looking at the group number. For Groups 13-18, subtract 10 to obtain the valence electrons
3. Place dots representing the valence electrons on the four sides of the element symbol singly before pairing up

only cares about
valence e^-





Checkpoint



Draw the Lewis Dot Diagram of:

a) Silicon



b) Ca



c) Mg^{2+}

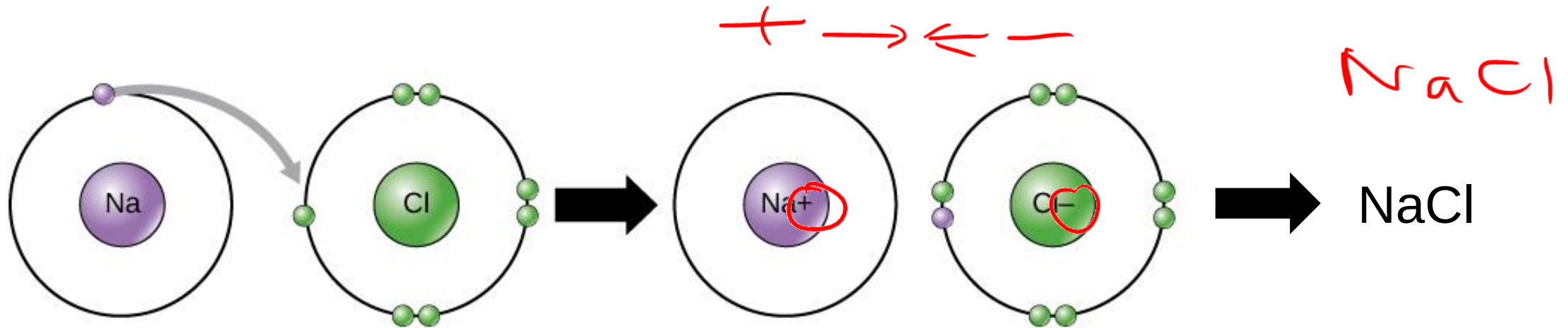
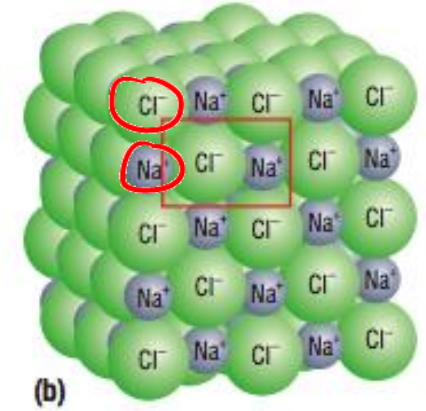
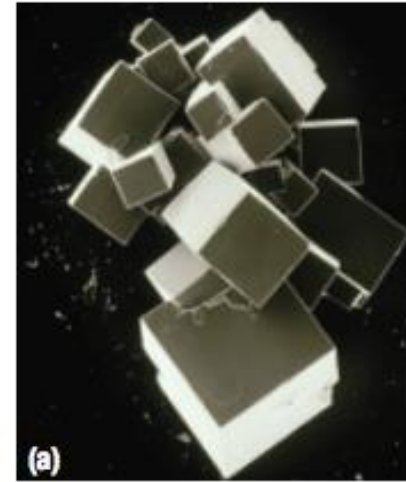


d) O^{2-}



Ionic Compounds

- **Ionic Compounds** are formed by the electrostatic interaction of a cation and an anion; metal and a nonmetal



exception H^+ usually

Periodic Table of the Elements

GROUP 1																		GROUP 18																		
1	1.01																	2	4.00																	
1	H	2.2																	18	He																
	Hydrogen																			Helium																
	+1, -1																																			
3	6.94	4	9.01															5	10.81	6	12.01	7	14.01	8	16.00	9	19.00	10	20.18							
2	Li	0.88	Be	1.67															13	B	2.04	14	C	2.55	15	N	3.04	16	O	3.44	17	F	3.98	Ne		
	Lithium		Beryllium																	Boron			Carbon		Nitrogen		Oxygen		Fluorine		Neon					
	+1		+2																	+3			-3		-2		-1									
11	22.99	12	24.31															13	Al	1.81	14	Si	1.90	15	P	2.19	16	S	2.58	17	Cl	3.16	18	Ar		
3	Na	0.93	Mg	1.31																Aluminum		Silicon		Phosphorus		Sulfur		Chlorine		Argon						
	+1		+2																	+3			-3		-2		-1									
19	39.10	20	40.08	21	44.96	22	47.87	23	50.94	24	52.00	25	54.94	26	55.85	27	58.93	28	58.69	29	63.55	30	65.41	31	69.72	32	72.64	33	74.92	34	78.96	35	79.90	36	83.80	
4	K	0.82	Ca	1.00	Sc	1.38	Ti	1.64	V	1.83	Cr	1.88	Mn	1.65	Fe	1.83	Co	1.88	Ni	1.81	Cu	1.80	Zn	1.85	Ga	1.81	Ge	2.01	As	2.18	Se	2.55	Br	2.98	Kr	
	Potassium		Calcium		Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton	
	+1		+2		+3		+4, +3		+5, +4		+3, +2		+2, +4		+3, +2		+2, +3		+2, +3		+2, +1		+2		+3		+4		-3		-2		-1			
37	85.47	38	87.62	39	88.91	40	91.22	41	92.91	42	95.94	43	98	44	101.07	45	102.91	46	106.42	47	107.87	48	112.41	49	114.82	50	118.71	51	121.76	52	127.60	53	126.90	54	131.29	
5	Rb	0.82	Sr	0.95	Y	1.22	Zr	1.83	Nb	1.80	Mo	2.16	Tc	1.9	Ru	2.20	Rh	2.28	Pd	2.20	Ag	1.83	Cd	1.88	In	1.78	Sn	1.98	Sb	2.05	Te	2.10	I	2.86	Xe	2.60
	Rubidium		Strontium		Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Rhodium		Palladium		Silver		Cadmium		Indium		Tin		Antimony		Tellurium		Iodine		Xenon	
	+1		+2		+3		+4		+5, +3		+6		+7		+3		+3		+2, +4		+1		+2		+3		+4, +2		+3, +5		-2		-1			
55	132.91	56	137.33	Lanthanide Series		72	178.49	73	180.95	74	183.84	75	186.21	76	190.23	77	192.22	78	195.08	79	196.97	80	200.59	81	204.38	82	207.2	83	208.98	84	209	85	210	86	222	
6	Cs	0.79	Ba	0.89		Hf	1.3	Ta	1.6	W	2.38	Re	1.9	Os	2.2	Ir	2.2	Pt	2.28	Au	2.64	Hg	2	Tl	1.82	Pb	2.33	Bi	2.02	Po	2.0	At	2.2	Rn		
	Cesium		Barium			Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Gold		Mercury		Thallium		Lead		Bismuth		Polonium		Astatine		Radon		
	+1		+2			+4		+5		+6		+7		+4		+4		+4, +2		+3, +1		+2, +1		+1, +3		+2, +4		+3, +5		+2, +4		-1				
87	223	88	226	Actinide Series		104	261	105	262	106	266	107	264	108	277	109	268	110	281	111	272	112	285	113	286	114	289	115	290	116	292	117	294	118	294	
7	Fr	0.7	Ra	0.8		Rf		Db		Sg		Bh		Hs		Mt		Ds		Rg		Cn		Nh		Fl		Mc		Lv		Ts		Og		
	Francium		Radium			Rutherfordium		Dubnium		Seaborgium		Bohrium		Hassium		Meitnerium		Darmstadtium		Roentgenium		Copernicium		Nihonium		Flerovium		Moscovium		Livermorium		Tennessine		Oganesson		
	+1		+2			+4		+5		+6		+7		+4		+4		+4, +2		+3, +1		+2, +1		+1, +3		+2, +4		+3, +5		+2, +4		-1				

57	138.91	58	140.12	59	140.91	60	144.24	61	145	62	150.36	63	151.96	64	157.25	65	158.93	66	162.50	67	164.93	68	167.26	69	168.93	70	173.04	71	174.97	
La	1.10	Ce	1.12	Pr	1.18	Nd	1.14	Pm		Sm	1.17	Eu		Gd	1.20	Tb		Dy	1.22	Ho	1.23	Er	1.24	Tm	1.25	Yb		Lu	1.27	
Lanthanum		Cerium		Praseodymium		Neodymium		Promethium		Samarium		Europium		Gadolinium		Terbium		Dysprosium		Holmium		Erbium		Thulium		Ytterbium		Lutetium		
+3		+3		+3		+3		+3		+3, +2		+3, +2		+3		+3		+3		+3		+3		+3		+3, +2		+3		
89	227	90	232.04	91	231.04	92	238.03	93	237	94	244	95	243	96	247	97	247	98	251	99	252	100	257	101	258	102	259	103	262	
Ac	1.1	Th	1.3	Pa	1.5	U	1.38	Np	1.38	Pu	1.28	Am	1.3	Cm	1.3	Bk	1.3	Cf	1.3	Es	1.3	Fm	1.3	Md	1.3	No	1.3	Lr		
Actinium		Thorium		Protactinium		Uranium		Neptunium		Plutonium		Americium		Curium		Berkelium		Californium		Einsteinium		Fermium		Mendelevium		Nobelium		Lawrencium		
+3		+4		+5, +4		+6, +4		+5		+4, +6		+3, +4		+3		+3, 4		+3		+3		+3		+3		+2, +3		+2, +3		+3

Metals tend to lose valence electrons

Non-metals tends to gain valence electrons

Physical Properties of Ionic Compounds

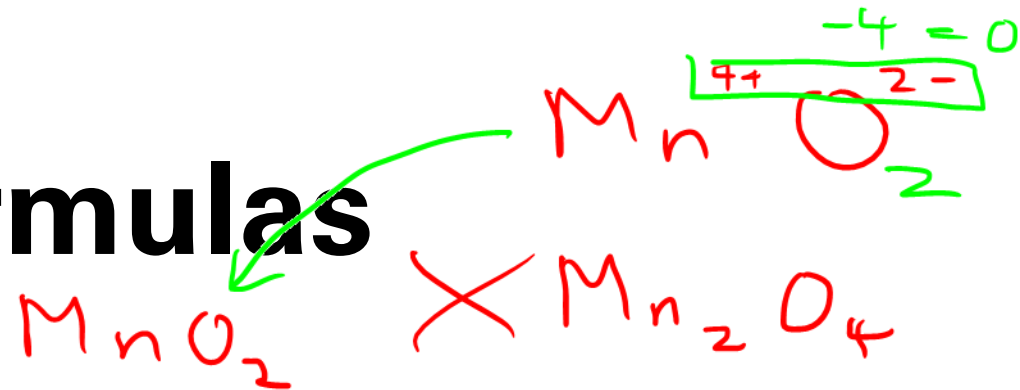
- High melting points and boiling points
- Low volatility/Non-volatile
- Generally soluble in polar solvents
- Do not conduct electricity in the solid state
- Conduct electricity when molten or when dissolved in water
- Generally brittle

Ionic compound	Charge on metal ion	Melting point
Na ₂ O	1+	1132 °C
MgO	2+	2800 °C

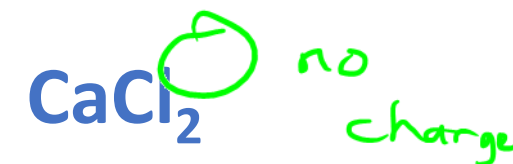
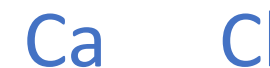
Writing Chemical Formulas

Crisscross Method

1. Write the element symbols of the metal and then the nonmetal
2. Find the ionic charge of each element on the periodic table and write the ionic charge on the top-right corner
3. Criss-cross the ionic charge
4. Simplify the subscripts to the lowest terms and remove the charges



Ex: Calcium chloride





Checkpoint



Draw the Lewis Dot Diagram for the formation of the following:

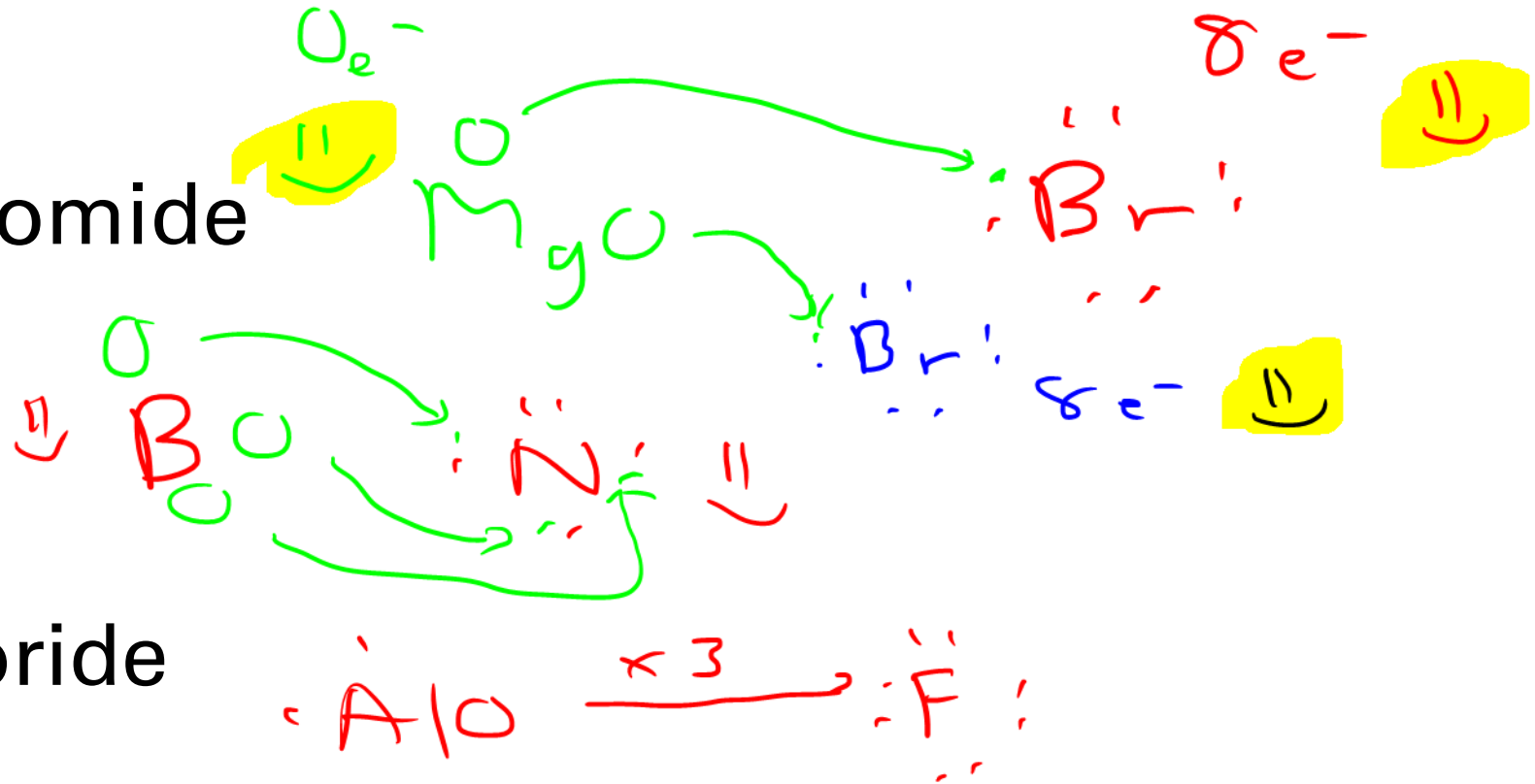
a) Magnesium bromide



b) Boron nitride



c) Aluminum fluoride



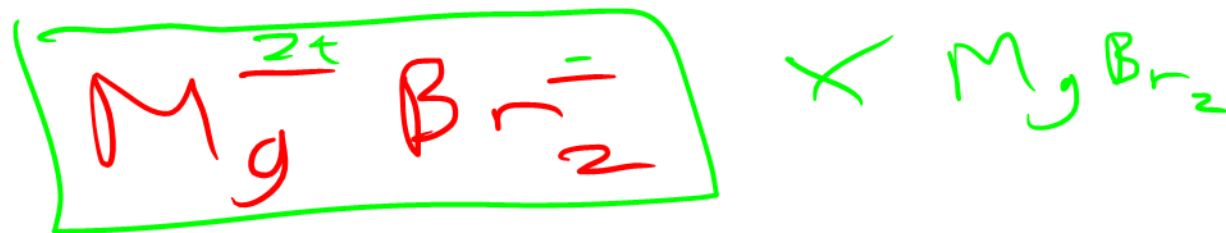


Checkpoint



Write the chemical formula for the following:

a) Magnesium bromide



b) Boron nitride



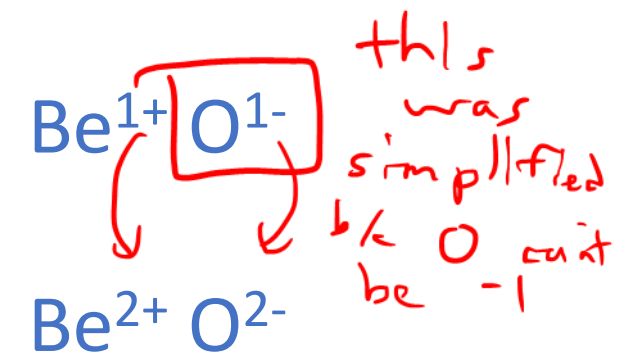
c) Aluminum fluoride



Writing Chemical Names

1. Un-crisscross the charges and consider if the compound was simplified. *check periodic table*
2. Write the name of the metal and check its ionic charge on the periodic table
3. Write the name of the nonmetal and check its ionic charge on the periodic table. For anions, change the ending to *-ide*

Ex: ~~BeO~~



Beryllium oxygen

Beryllium oxide

Multivalent Elements

- Some transition metals are multivalent (more than one ionic charge)
- Use Roman numerals to distinguish between the ionic charges

1A	2A											13A	14A	15A	16A	17A	8A
Li ⁺												Al ³⁺	C ⁴⁺	N ³⁻	O ²⁻	F ⁻	
Na ⁺	Mg ²⁺	3B	4B	5B	6B	7B	8B	9B	10B	11B	12B			P ³⁻	S ²⁻	Cl ⁻	
K ⁺	Ca ²⁺				Cr ²⁺ Cr ³⁺	Mn ²⁺ Mn ³⁺	Fe ²⁺ Fe ³⁺	Co ²⁺ Co ³⁺	Ni ²⁺ Ni ³⁺	Cu ⁺ Cu ²⁺	Zn ²⁺				Se ²⁻	Br ⁻	
Rb ⁺	Sr ²⁺									Ag ⁺	Cd ²⁺		Sn ²⁺ Sn ⁴⁺		Te ²⁻	I ⁻	
Cs ⁺	Ba ²⁺									Au ⁺ Au ³⁺	Hg ₂ ²⁺ Hg ²⁺		Pb ²⁺ Pb ⁴⁺				

Ion	Chemical Name
Fe ²⁺	Iron(II)
Fe ³⁺	Iron(III)
Cu ⁺	Copper(I)
Cu ²⁺	Copper(II)



Checkpoint



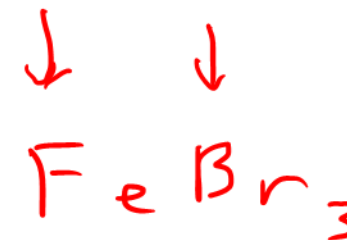
Write the chemical formula for the following:

FeBr_3 X
 FeBr_3 ✓

a) Copper(II) chloride



b) Iron(III) bromide



c) Nickel(II) oxide





Checkpoint



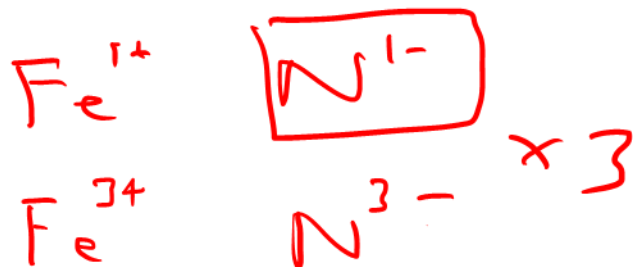
Write the chemical name for the following:



Iron (III) oxide

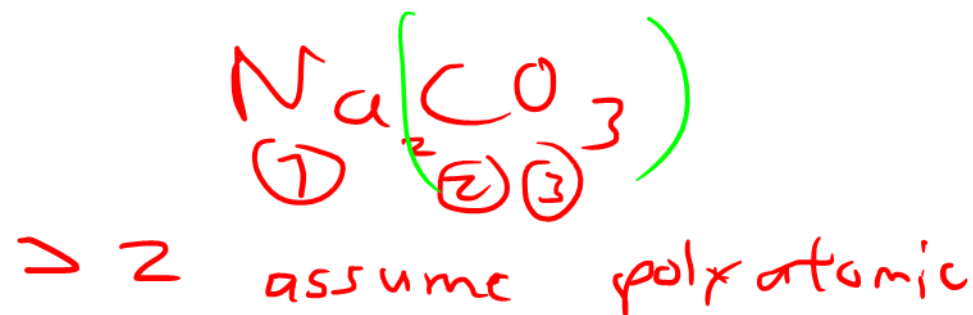


Copper (I) sulfide ✓
sulphide ✓



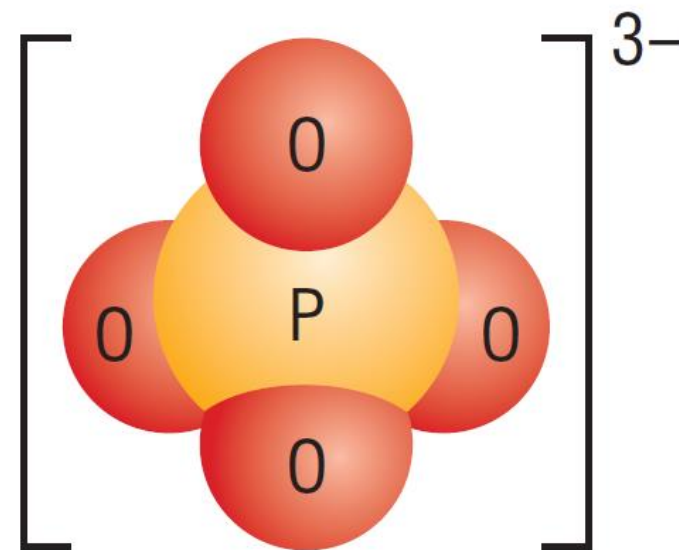
Iron (III) nitride

Polyatomic Ions



- **Polyatomic ion** is made of more than one atom and acts as a single unit

Ex: PO_4^{3-} phosphate ion
 OH^- hydroxide ion
 CO_3^{2-} carbonate ion



TIP: It may be helpful to keep the elements within the parentheses and the charge outside the parentheses.

Ex: $(\text{PO}_4)^{3-}$ or $(\text{OH})^-$

Writing Chemical Formulas with Polyatomic Ions

same as solving for NaCl

1. Write the metal with its ionic charge

Ex: Iron(III) nitrate

2. Write the polyatomic ion with its charge outside the parentheses



3. Crisscross the ionic charges



4. Simplify the subscripts



- Do not simplify the elements within the parentheses

Naming Chemical Names with Polyatomic Ions

1. Place parentheses around the polyatomic ion



2. Un-crisscross the charges



3. Write the name of the metal and check its ionic charge on the periodic table



4. Write the name of the polyatomic ion and check its ionic charge

from pg 2 of periodic table

Sodium carbonate



Checkpoint



doesn't end in -ide = polyatomic

Write the chemical formula for the following:

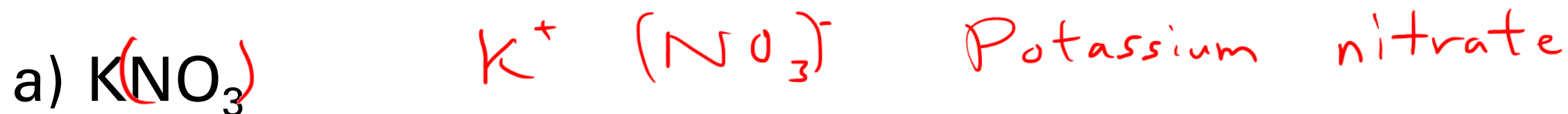
- a) Calcium sulfate $\text{Ca}^{2+} (\text{SO}_4)^{2-}$ CaSO_4
- b) Calcium chlorate $\text{Ca}^{2+} (\text{ClO}_3)^{-}$ $\text{Ca}(\text{ClO}_3)_2$
- c) Iron(III) phosphate $\text{Fe}^{3+} (\text{PO}_4)^{3-}$ FePO_4



Checkpoint

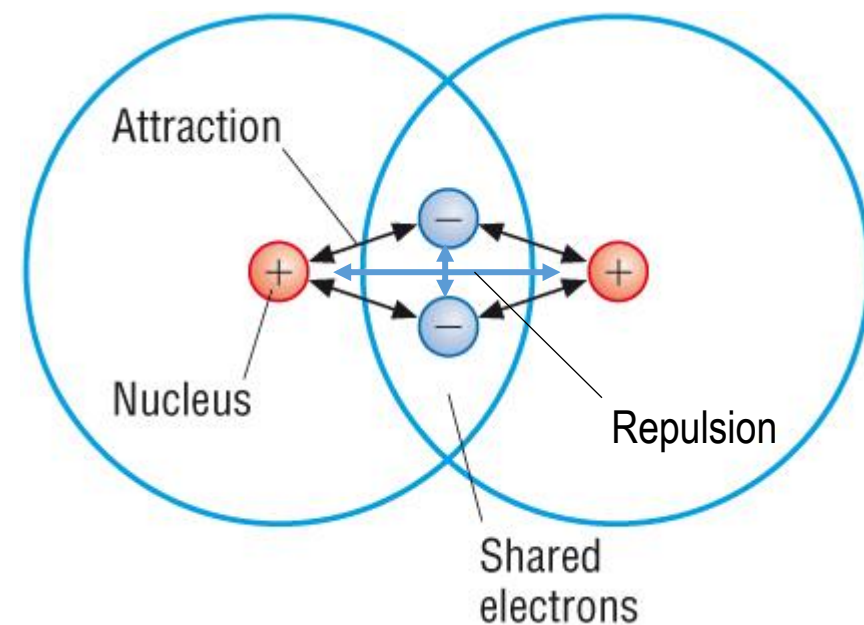


Write the chemical name for the following:



Covalent Bonding

- **Covalent bonding** is the stable balance of attractive forces and repulsive forces between atoms
 - Formed between nonmetals



a) Nitrous oxide (N_2O) is commonly known as laughing gas



b) Nitrogen dioxide (NO_2) is the reddish-brown gas in smog

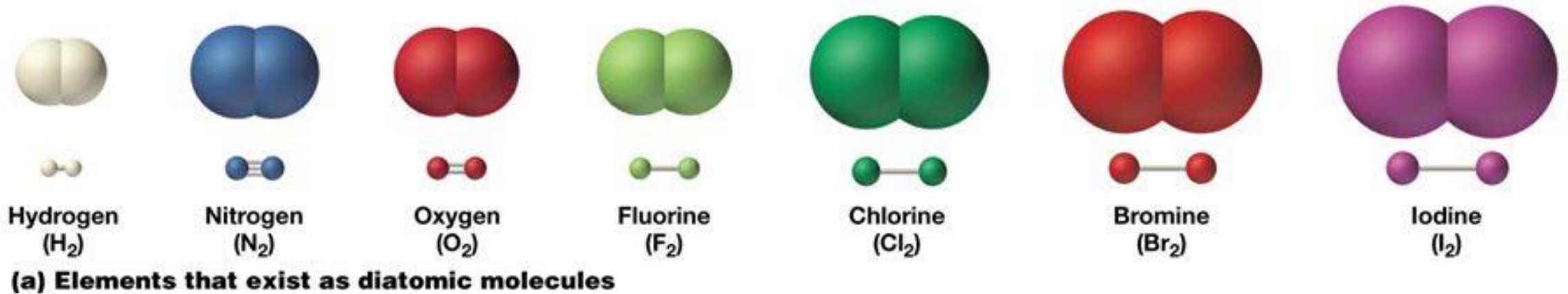
Physical Properties of Covalent Compounds

- Lower melting points and boiling points than ionic compounds
- Some are volatile (i.e. F_2 , Cl_2 , Br_2)
- Low solubility in water
- Do not usually conduct electricity as a solid, liquid, or in the aqueous state



Diatomic Molecules

- **Diatomic molecules** are molecules that consist of two atoms joined with covalent bonds
- Remember: H_2 O_2 F_2 Br_2 I_2 N_2 Cl_2 *Also P_4 and S_8



sodium oxide

Naming Covalent Molecules

1. Write the name of the elements
2. Change the ending of the second element to *-ide*
3. Add prefixes to represent the number of atoms
 - If mono- is the first prefix, it is understood and not written
 - If the element starts with a vowel, drop the (a) on the prefix

Ex: CO₂

carbon oxygen

carbon oxide

carbon dioxide

Number	Prefix
1	Mon(o)-
2	Di-
3	Tri-
4	Tetr(a)-
5	Pent(a)-
6	Hex(a)-
7	Hept(a)-
8	Oct(a)-
9	Non(a)-
10	Dec(a)-



Checkpoint



Write the chemical names for the following:

a) CO *carbon monoxide*

b) PF_5

c) N_2O

Writing Molecular Formulas

1. Write the element symbol
2. Refer to the prefix and add subscripts to the element symbol
3. Do not simplify for covalent molecules

Ex: sulfur dioxide

S O

S₁ O₂

SO₂



Checkpoint



Write the chemical formulas for the following:

a) Sulfur tetroxide

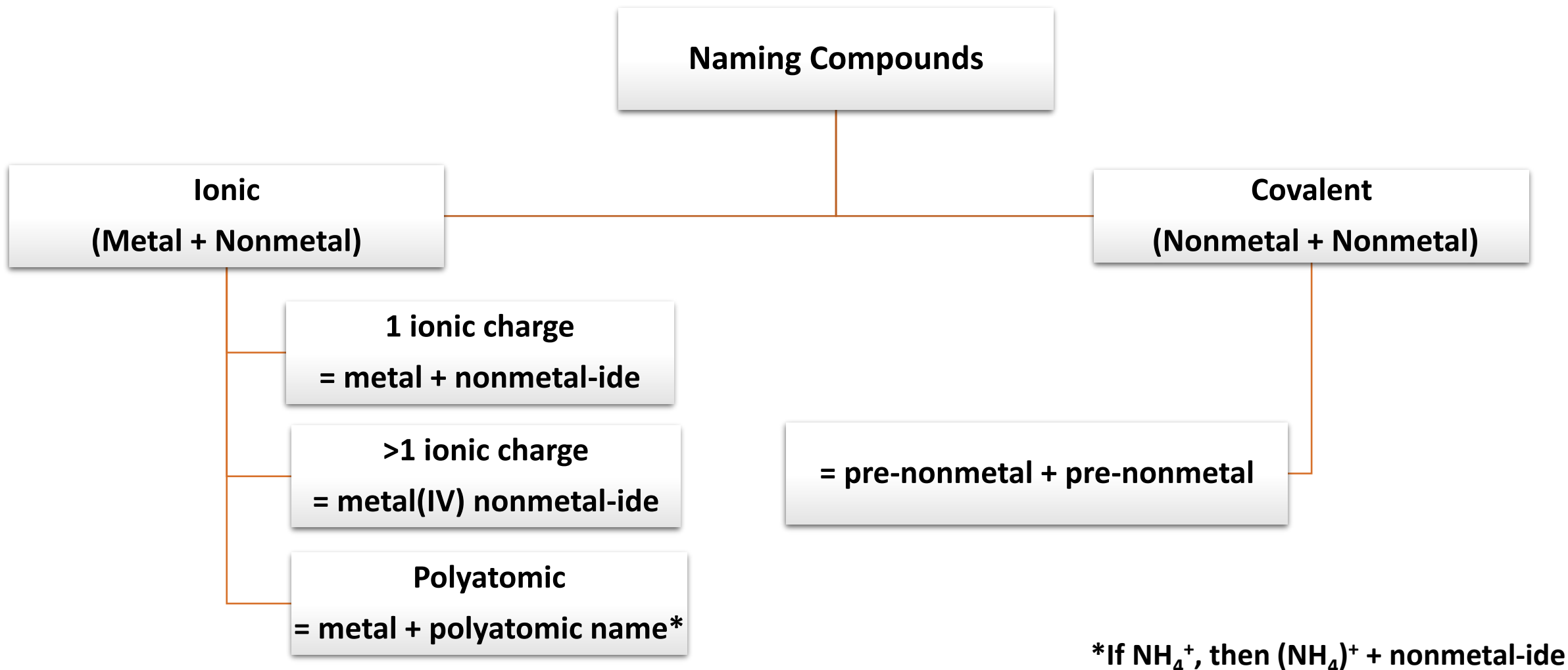


b) Dihydrogen dioxide



c) Carbon disulfide

Summary: Nomenclature



What I Learned Today:

- ☐ Lewis Dot Diagrams
- ☐ Physical Properties of Ionic Compounds
- ☐ Naming and Writing Ionic Compounds with:
 - Single charge
 - Multivalent charges
 - Polyatomic ions
- ☐ Naming and Writing Covalent Molecules

Due next class: Class 1 Homework

